

RAG-based cybersecurity assistant — build workflow

Retrieval pipeline through deployment, end to end

1. Data sources

CVE/NVD feeds, MITRE ATT&CK, ICS-CERT advisories, internal SOPs



2. Preprocessing & chunking

Clean sources, normalize CVE/CVSS fields, chunk by type, tag metadata



3. Embedding & indexing

Generate embeddings, build hybrid vector + keyword index



4. Retrieval & re-ranking

Metadata filtering, hybrid search, cross-encoder re-rank



5. Grounded generation

LLM answers strictly from retrieved chunks, with citations back to source



6. Guardrails & safety

Refuse exploit/attack instructions, flag low-confidence answers



7. Evaluation

Golden test set — retrieval precision/recall + generation faithfulness



8. Deployment & monitoring

API/UI layer, query + retrieval logging, drift and cost tracking

